

## source free domain adaptation

یادداشت

- Issues: 1- maintain large databases 2- hard
- 2- confidentiality & Privacy
- \* UDA needs source data but sFDA train the model on source data and completely foregoes and adapt only using unlabeled target data
- \* how align distribution or measure gap between source & target? adapt with no label could cause model drift & accuracy plummet (how make sure improving and not breaking)
- \* 2 path for solution: **model driven** & **data driven**
- \* **model driven** update model by self supervised learning (rely on model's output on target data) but **data driven** look outward and absence of source data, like reconstruct it or get deep in target data structure
- \* **model driven**: use model <sup>and</sup> choose label of model on target data with high confident and train the model with them
- ↳ it causes model reinforce its mistakes

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which is central challenge of pseudo labeling  
(very dependent to how handle noise  $\rightarrow$  potential  
incorrect label)

$\hookrightarrow$  how it's handled: 1- what each class look like  
in feature space, often use weights of classifier  
Part of model as starting point of class centers  
(Prototype) which is using model's existing ~~knowledge~~  
knowledge and <sup>2-</sup> assign target samples to nearest  
pseudo labels of nearest class prototype  
(<sup>or</sup> ~~the~~ class that model predict with highest confidence  
score) 3- filter and reject labels that are likely  
wrong like compare the closest sample of assign  
class prototype by next closest one or ~~stb~~ <sup>looking</sup>  
for stability (does predicted class changing  
during adaptation)

another method (entropy minimize): adding extra  
objective which measure uncertainty  $\rightarrow$  well  
adapted, ~~model~~ <sup>model</sup> should produce certain  
Prediction for each sample (low uncertainty)



means entropy) so add loss for reduce entropy, but cause cheating model by choosing easiest ~~also~~ class for everything which is certain but useless, to address this issue they used information maximization loss which make low entropy for individual prediction certainty but want high entropy across batch of prediction diversity (confidently use all possible classes) ~~selecting~~ another method (contrastive learning): needs pair of positive & negative, they used model classification weights acting as prototype for each class, so we try to pull the target samples ~~to~~ the features closer to its assigned prototype, push further from other classes prototypes (no need for source data sample)

data driven:

- domain based reconstruction generate source data (or close enough)





1. using generative models like gen. ANN, some methods like 3c-gen make image not only look real but also match meaning of class that it going to represent
2. implicit knowledge generation: trained model got some statistical data of source, for example, batch normalization (BN) layers that store  $\mu, \sigma$  of feature activation of source batches and used gaussian distribution parameterized by  $\mu, \sigma$  to generate virtual source features (with out needing original img or heavy generation)
3. intra domain adversarial alignment: use <sup>source</sup> ~~some~~ model prediction to split target samples to high confident <sup>(more</sup> ~~and~~ source like) and low confident (more different & target specific) and make adversarial game in target domain  $\rightarrow$  train feature extractor to make low confident sample look



like high confident ones (Pushing model decision boundary to better accomodate target specific samples and shrink target specific group) make model more robust to target specific samples

\* ~~target~~ based info extraction: neighbourhood clustering  $\rightarrow$  even domains are different, samples from same class should still cluster together in feature space  $\rightarrow$  samples which are close neighbour in feature space to have consistent labels like  $CR5+80 \rightarrow$  clean up decision boundary between clusters is effective

ing style transition: source model has confused by target data style, so make style source like style, but main content preserve  $\rightarrow$  it uses BN layers statistical signature of source style, so make target samples feature stats match source BN stats

\* self attention: helps model to focus nicholding.ir



what to focus on

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on the most important part of img (actual obj) and ignore background clutter and style change ~~etc~~ which are usually source of domain shift  $\rightarrow$  models like dsitB

\* no solid general theory has ~~std~~ and more is based on intuition & <sup>emp</sup> ~~emp~~irical also we need more benchmark (complex, class imbalance, noisy target domain)